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Broad Trends in Geomagnetic Paleointensity on Hawaii During Holocene Time

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Paleointensity determinations have been obtained from 22 basaltic lava flows on the island of Hawaii using the Thelliers' method. Radiocarbon dating indicates that these flows erupted at intervals ranging from about 200 to 1000 years, and results of the experiments provide an estimate of broad trends in geomagnetic paleointensity during Holocene time in the vicinity of Hawaii. Most of the samples were obtained from quickly cooled flow margins and, as a consequence, typically contain two titanomagnetite populations. Only two of the 79 samples analyzed failed to yield an estimate of the paleofield, demonstrating that paleointensities can be obtained from such samples if carefully selected. Virtual dipole moments calculated for the flows are compared with a published curve of dipole field intensity that was constructed using worldwide archeomagnetic data. The large nondipole fields that were previously postulated for the vicinity of Hawaii are confirmed, and the present data indicate that they were present from about 5000 years B.P. to perhaps as recently as the past 200 years. The data indicate, however, that these nondipole sources must have been virtually absent between about 12,000 and 5000 years B.P. as they are at the present time.

INTRODUCTION

Paleomagnetic samples have been obtained by the U.S. Geological Survey from several hundred lava flows erupted from Kilauea, Mauna Loa, and Hualalai volcanoes on the island of Hawaii. The most recent studies were initiated in an attempt to use the history of geomagnetic secular variation recorded by the lava flows to facilitate geologic mapping of the island and thereby aid in the assessment of volcanic hazards [*McWilliams et al.*, 1982; *Holcomb et al.*, 1986]. Within this large inventory of sampled flows are many that have been dated by the radiocarbon (^{14}C) method. We initiated a study of some of the ^{14}C-dated lava flows in order to characterize the variations of geomagnetic intensity throughout Holocene time and to use this information in combination with magnetic directions to enhance the utility of paleomagnetic secular variation as a correlation tool.

Several paleointensity studies have been conducted on lava flows on the island of Hawaii, the earliest being performed on samples from the Kau and Puna Volcanic Series [*Doell and Cox*, 1972]. Because the results of Doell and Cox were obtained using the *Wilson* [1961] method as modified by *Doell and Smith* [1969], they are now thought to be unreliable. (See the discussion by *Coe et al.* [1978].) *Abokhodair* [1977] studied samples from 16 historic lava flows in order to assess the accuracy of the *Thellier and Thellier* [1959] paleointensity method, as applied to these rocks, and obtained results that were accurate to within about 7% of the known field intensities. His results also indicated that this technique was more effective when the experiments were performed in vacuum [*Khodair and Coe*, 1975; *Abokhodair*, 1977]. *Coe et al.* [1978] used the Thelliers' method and obtained reliable paleointensities for samples from eight of the nine radiocarbon-dated lava flows that were sampled. The Thelliers' method also was used by *Tanaka and Kono* [1991] in their study of historic and ^{14}C-dated lava flows from Hawaii.

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RADIOCARBON AGES

Radiocarbon ages for many of the Hawaiian lava flows have been reported by *Kelley et al.* [1979] and *Rubin et al.* [1987*a*]. The flows studied during the current investigation were chosen to provide a sampling of the geomagnetic field at fairly regular intervals throughout Holocene time. Lava flows that ultimately were found to have magnetic properties indicating suitability for paleointensity experiments (see below) are listed in Table 1.

It has been shown [*International Study Group*, 1982] that variations in excess of quoted errors can occur in ^{14}C dating. The quoted errors usually are based only on counting statistics, whereas the true error is influenced by isotopic fractionation effects, contamination, and biases due to equipment and methods. These factors and the lack of both interlaboratory comparisons and a universally accepted method of error calculation make it difficult to compare results from different laboratories. Consequently, we follow the recommendations of the study group and assign uncertainties in Table 1 that are three times the quoted error for each determination.

The age of the Visitor Center flow was considered by *Coe et al.* [1978] to be within the ±250-year uncertainty about the 2570 years B.P. minimum age reported by *Rubin and Suess* [1956]. Subsequently, J. P. Lockwood (written communication, 1983) reported that the age of this flow was probably in error due to contamination by "dead" CO_2 being emitted by fumaroles in the immediate vicinity. (See, for example, *Rubin et al.* [1987*b*].) Lockwood considered the actual age of the flow to be probably between 400 and 700 years old. *Holcomb et al.* [1986] also questioned the reported age and considered the flow to probably be younger than 500 years old.

The Panaewa flow, which is exposed in the Allied Aggregates quarry southeast of Hilo (site 10, Figure 1), originally yielded a ^{14}C age of 2890 ± 70 years B.P. (sample W4174 of *Kelley et al.* [1979]). Because this flow appears to be one of the youngest prehistoric flows in the Hilo quadrangle, Kelley et al. considered this age to be much too old. A subsequent determination on sample W4981 yielded a much younger age

TABLE 1. Paleomagnetic and Age Data for Hawaiian Holocene Lava Flows

Map (Figure 1)	Age ± 3σ, years B.P.	^{14}C Number	Site			N/N_0	H, mT	I, deg	D, deg	k	α_{95}, deg	VGP	
			Field	Latitude	Longitude							Latitude	Longitude
1a			8B793	19.545	204.912	12/12	15	41.3	5.5	630	1.7	83.4	254.6
1b	260 ± 210	W3881	9B121	19.580	204.938	12/12	20	44.2	3.2	906	1.4	83.0	229.2
1c			9B433	19.572	204.941	12/12	20	46.0	3.1	418	2.1	81.7	224.1
2	440 ± 270	W3790 W3793	8B829	19.552	204.698	12/12	20	36.6	7.6	1662	1.1	82.9	287.4
3	700 ± 210	W4402	1B468	19.296	204.689	18/20	20	41.4	2.8	905	1.2	84.8	234.3
4	760 ± 210	W4345	1B752	19.494	204.656	11/12	20	25.8	353.0	849	1.6	81.1	074.5
5	1,320 ± 150	W4237	9B109	19.564	204.656	12/12	20	18.2	2.2	400	2.2	79.6	012.8
6	1,690 ± 210	W4344	1B728	19.533	204.648	10/12	30	17.8	355.5	679	1.9	78.7	048.0
7	2,180 ± 180	W4015	8B889	19.118	204.426	12/12	10	14.7	11.2	439	2.1	74.1	340.0
8	2,190 ± 210	W3876	8B625	19.455	204.707	12/12	20	15.2	11.3	1095	1.3	74.0	340.0
9	2,550 ± 240	W4810	1B704	19.538	204.655	11/12	30	21.0	1.2	987	1.5	81.3	016.9
10	2,890 ± 210	W4174	9B037	19.640	204.958	12/12	20	25.7	357.7	370	2.3	83.5	044.9
11	3,480 ± 240	W3831	8B409	19.302	204.694	11/12	60	36.4	3.7	1091	1.4	86.4	279.5
12	4,050 ± 150	W3803	8B433	19.713	204.882	12/12	15	33.4	4.5	1467	1.1	85.5	313.1
13	5,160 ± 300	W4135	9B445	19.161	204.458	12/12	20	27.6	4.7	493	2.0	83.6	339.2
14	5,650 ± 270	W3862	8B901	19.543	204.876	12/12	20	33.7	5.4	958	1.4	84.8	305.8
15	6,160 ± 330	W3930	9B649	19.086	204.406	10/12	30	54.9	357.8	819	1.7	73.5	198.1
16	7,300 ± 300	W4117	9B961	19.072	204.443	12/12	30	25.4	2.3	914	1.4	83.9	003.0
17	7,950 ± 330	W3813	9B661	19.145	204.484	11/12	30	33.4	359.5	917	1.5	89.0	053.1
18	8,740 ± 300	W4803	9B937	19.420	204.661	11/12	40	36.0	357.3	413	2.3	87.4	127.2
19	9,500 ± 420	W3840	9B949	19.418	204.656	12/12	40	33.7	3.0	351	2.3	87.0	313.6
20	10,290 ± 450	W4160	9B481	19.101	204.441	12/12	30	31.9	3.6	570	1.8	86.1	322.0
21	11,780 ± 300	W3487	9B901	18.972	204.378	11/12	40	8.7	2.1	738	1.7	75.2	016.2
22	13,210 ± 570	W4121	9B973	19.072	204.443	12/12	30	54.6	357.5	916	1.4	73.8	197.2

Radiocarbon age and sample number from *Rubin et al.* [1987*a*]; Latitude and longitude are north latitude and east longitude, respectively, of paleomagnetic sampling site and virtual geomagnetic pole (VGP); N/N_0 is the number of samples used in the calculation of mean direction/total number of samples collected; H is the mean demagnetizing field; I is inclination, positive downward; D is declination, east of north; k is the precision parameter of *Fisher* [1953]; α_{95} is the 95% confidence cone about average direction.

of 1470 ± 50 years B.P. [*Rubin et al.*, 1987*a*]. The remanence direction for this flow (Table 1), however, seems more consistent with the older age as shown by *Holcomb et al.* [1986], and we follow that interpretation here. Additional age determinations probably will be necessary to resolve this discrepancy.

Methods

Paleomagnetic samples were obtained from individual lava flows at the localities shown in Figure 1. A minimum of 12 cores was collected from each flow using a portable core drill and oriented with a sun compass. The remanent magnetization of one specimen from each core was measured using either a spinner or cryogenic magnetometer. Two representative specimens from each lava flow were subjected to progressive alternating field (AF) demagnetization to test the stability of remanence. One specimen from each of the remaining cores in a flow was then magnetically cleaned at a field strength that was chosen on the basis of behavior during the progressive demagnetization experiments. The remanent magnetization directions, after magnetic cleaning, for all units studied are given in Table 1.

Strong field thermomagnetic measurements were performed in vacuum using an automatic recording balance. Samples were heated and cooled at a rate of 10°C/min in applied fields between 200 and 500 mT. Curie temperatures (T_c) were measured using the method of *Grommé et al.* [1969]. Saturation magnetization (J_s) and saturation remanence (J_{rs}) were determined for the same samples later used in the strong field thermomagnetic measurements. J_{rs} was measured after magnetizing the samples in 900 mT, and J_s was determined by plotting the variation of the induced magnetization (in fields ranging from 0 to 900 mT) against the reciprocal of the applied field and extrapolating to $H = \infty$. Weak field susceptibilities were measured at 800 Hz using a commercial bridge at room temperature.

Sample Selection Criteria

Specimens used in the paleointensity experiments were selected primarily on the basis of their behavior during strong field thermomagnetic ($J_s - T$) analysis. A second requirement was that the specimen selected should have a remanence direction close to the mean direction for the flow in order to avoid samples that contain spurious components of magnetization. Because the Hawaiian lava flows are young and, generally, uneroded or unfaulted, paleomagnetic sampling was confined mainly to quickly cooled flow margins. Occasionally, cooling of the lava was too rapid to allow high-temperature oxidation of the magnetic oxides and $J_s - T$ curves, in these cases, showed only a single, low to intermediate Curie temperature that is indicative of a primary high-titanium titanomagnetite (Figures 2*a* and 2*b*). More commonly, incomplete high-temperature oxidation of a primary high-Ti titanomagnetite to a low-Ti titanomagnetite containing ilmenite lamellae resulted in $J_s - T$ curves showing two Curie temperatures (Figures 2*c*–2*f*). Almost as rare as the unoxidized samples were those that cooled slowly enough to have undergone complete high-temperature oxidation (Figure 2*g* and 2*h*). Moreover, some of the large variations in magnetic mineralogy often occurred within a single core (Figure 3) and indicated the necessity of perform-

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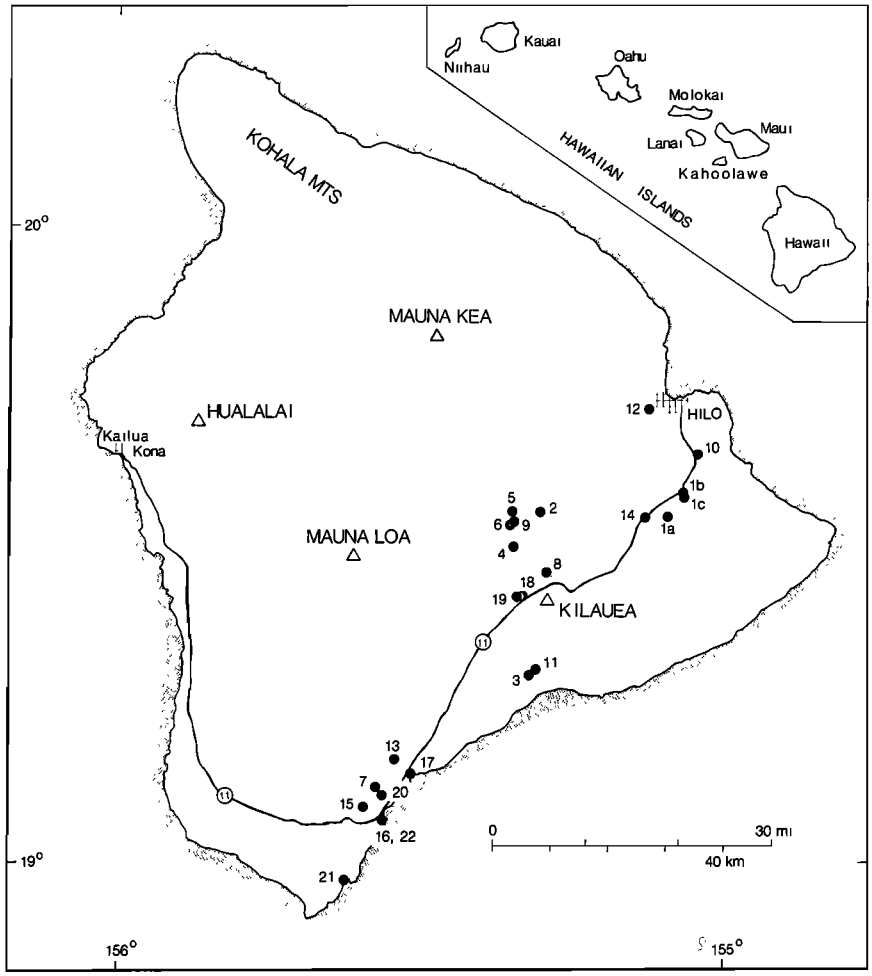


Fig. 1. Map of the island of Hawaii showing paleomagnetic sample localities. Numbers correspond to those given in Table 1.

ing $J_s - T$ and paleointensity experiments on the same 2.54-cm-long specimen.

Although two Curie temperature samples have been known to be difficult to use for paleointensity determinations [e.g., *Ade-Hall et al.*, 1968; *Coe et al.*, 1984], we adopted two requirements that we hoped would minimize the difficulties that are sometimes encountered. First, greater than 50% of the magnetization indicated by the $J_s - T$ curve should reside in the low-temperature component in order to obtain a significant fraction of the total natural remanent magnetization (NRM) intensity to interpret, and, second, the two Curie temperatures should be as distinct as possible. We reasoned that a sample showing a $J_s - T$ curve such as that in Figure 2*c* would not contain a broad continuum of magnetic minerals that were oxidized to varying degrees but would have two discrete titanomagnetite populations. Such ideal samples, of course, were difficult to find, and the samples that were ultimately selected had differing proportions of the two populations (Figures 2*c*–2*e*) and/or low-Curie-temperature material that was much more distributed (Figure 2*f*).

Other magnetic property measurements were made on several samples from each flow in order to determine if additional criteria could be used to select samples that would provide reliable paleointensity estimates; these are summarized in Table 2. It was found that the paleomagnetic sampling had produced a sample population whose magnetic properties were largely homogeneous. The geometric mean ratio of J_{rs} to J_s for the samples chosen for the paleointensity experiments (shaded area, Figure 4*a*) was found to be 0.20 and values between 0.30 and 0.13 are within one standard deviation of this mean. This ratio is nearly identical to that of all the samples that were measured, where J_{rs}/J_s was determined to be 0.18 with a 1 s.d. range between 0.29 and 0.12. These values are well within the $0.17 < J_{rs}/J_s < 0.27$ range that *Dunlop* [1983] determined for a group of basalt samples that displayed single-domain (SD) behavior. Only a small part of the sample population yields a J_{rs}/J_s near 0.1, which is indicative of multidomain (MD) to pseudo-single-domain (PSD) grains. That most of the samples contain single-domain titanomagnetite is not surprising because many of the cores were obtained from the glassy, quickly cooled margins of the lava flows. However, J_{rs}/J_s does not differ appreciably for those samples that were cooled more slowly as attested to by a single, high Curie temperature in the $J_s - T$ curves. The high-temperature oxidation indicated by these curves represents formation of magnetitite/ilmenite intergrowths where uniaxial shape anisotropy is prevalent for the magnetite particles. Although a random distribution of SD grains having uniaxial anisotropy should have a J_{rs}/J_s

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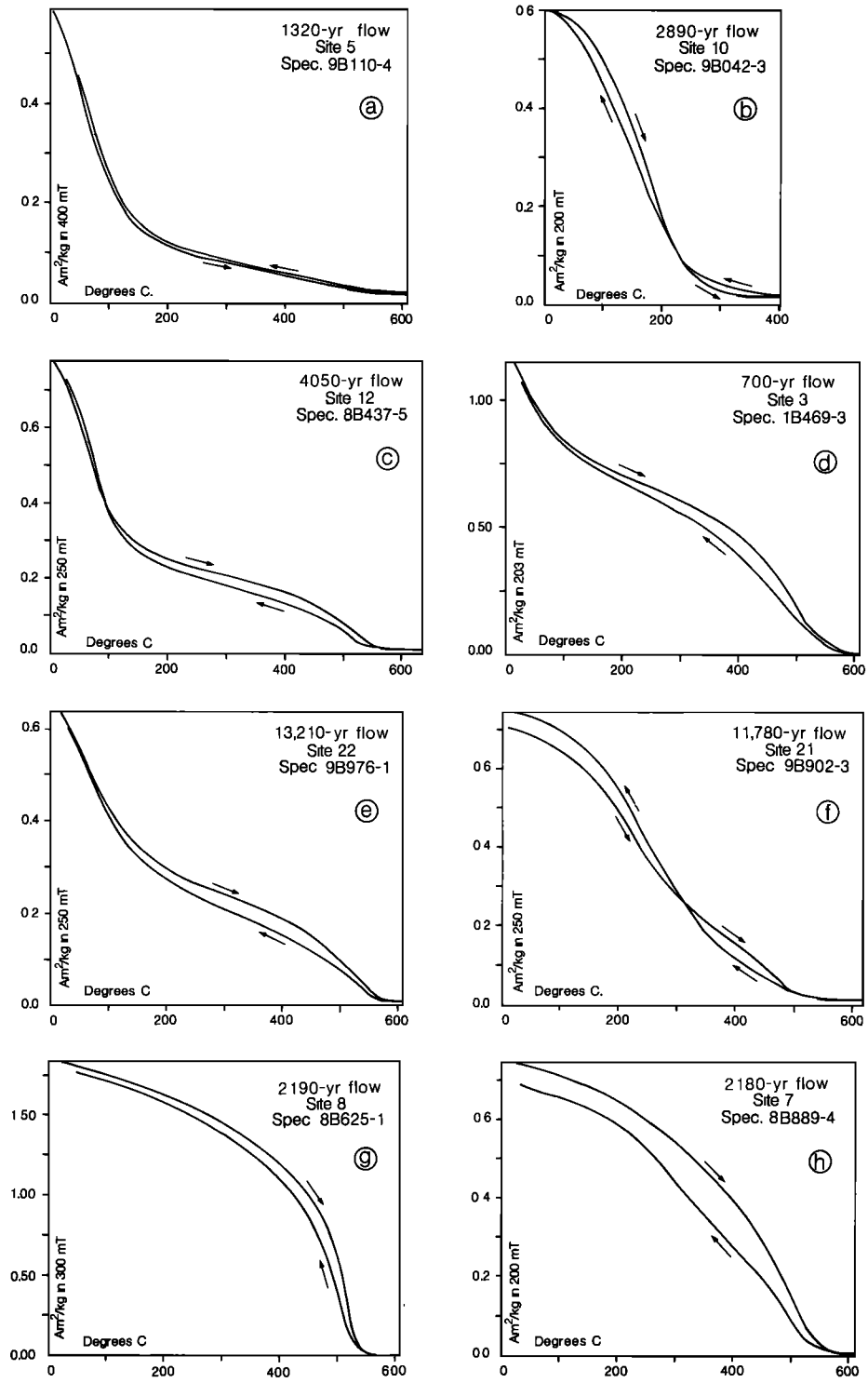


Fig. 2. Selected strong field thermomagnetic curves for samples from the Hawaiian ^{14}C-dated lava flows. Heating and cooling curves indicated by arrows. Experiments were run in vacuum at heating and cooling rates of 10°C/min.

ratio of 0.5, *Davis and Evans* [1976] show that this ratio is reduced to 0.3 because of strong interactions between SD units within the composite titanomagnetite grain.

An in situ Koenigsberger ratio Q_n was calculated for each of 301 samples, and these yielded a geometric mean of 46 (Figure 4*b*). Values between 24 and 86 are within one standard deviation of this mean. Because the Koenigsberger ratio is dependent on geomagnetic field strength as well as grain size of the magnetic minerals, we also calculated a laboratory Koenigsberger ratio Q_l [*Prévot et al.*, 1985], using the weighted mean paleointensity (Table 3) for each flow. These calculations yielded a geometric mean Q_l of 39 (Figure 4*c*), and values between 20 and 76 are within one standard deviation. Koenigsberger ratios as high as these are indicative of SD particles, as is a geometric mean, median destructive field of 30 mT (s.d. range 20 to 50) for represen-

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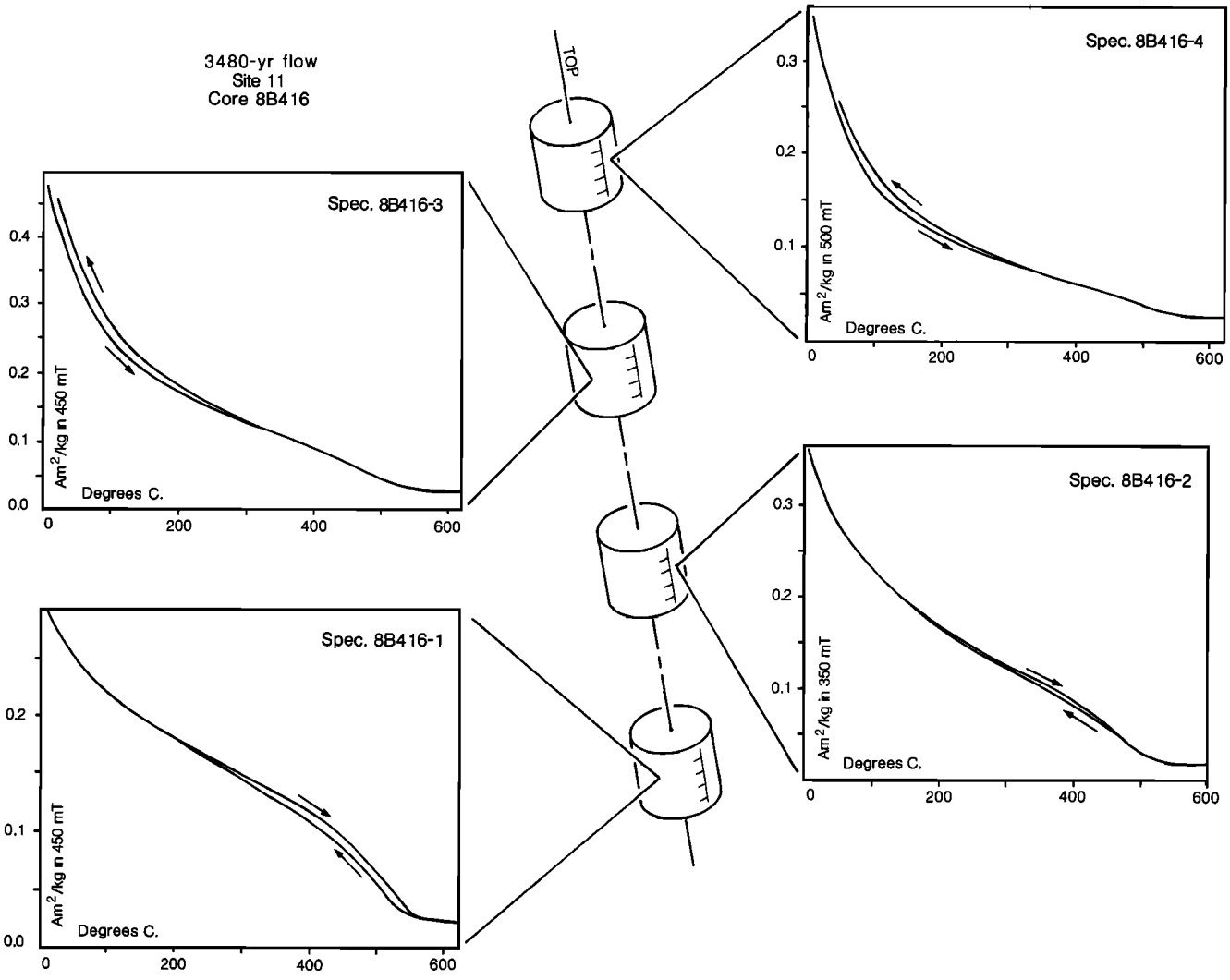


Fig. 3. Strong field thermomagnetic curves for specimens from a single core from the 3480 year B.P. flow (site 11).

tative samples from the lava flows studied. There seems little doubt that our samples primarily contain single-domain particles which are ideal for paleointensity experiments [e.g., *Levi*, 1977].

Paleointensity Experiments

The paleointensity experiments were performed in vacuum using the apparatus described by *Prévot et al.* [1985]. The ambient field within the furnace array was tested before beginning each new sequence of experiments to ensure that a minimal field was maintained. The field measured at each sample position ranged from 20 to 170 nT, with an average of about 70 nT for each set of experiments. We used the paleointensity method developed by *Thellier and Thellier* [1959] as modified by *Coe* [1967]. At every temperature, the specimens were first heated and cooled in a "zero" field (the ambient field), and the remaining natural remanent magnetization (NRM) was measured at room temperature using a spinner or cryogenic magnetometer. The same magnetometer was used for all measurements on an individual specimen in order to eliminate possible errors arising from slight differences in instrument calibration. A field of 35 μT, which is close to the present field intensity in Hawaii, was applied parallel to the cylindrical axes of the specimens during the second of each pair of heating steps. This field was applied throughout the entire heating and cooling cycle [*Levi*, 1975]. Even though the Thelliers' method is applicable over a wide range of paleointensities, *Tanaka and Kono* [1984] have shown that experimental errors can be reduced by matching the laboratory field to the ancient field. However, because our initial set of experiments indicated that the field intensity over the age range being investigated varied from the present-day field by less than a factor of 2, errors arising from mismatched fields probably are minimal, and the same applied field was used throughout this study.

Heating times typically ranged from 1.5 to 2 hours, and the samples remained in the oven and were allowed to cool naturally after the predetermined temperature was reached. Thus each heating/cooling cycle was several hours long. Although some laboratories are employing short cooling times for their paleointensity experiments, it has been shown [*Fox and Aitken*, 1980; *Dodson and McClelland-Brown*, 1980; *Halgedahl et al.*, 1980] that the thermoremanent magnetization (TRM) acquired during slow cooling can be

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TABLE 2. Magnetic Properties of Samples Used for Paleointensity Determinations

Map (Figure 1)	Age, years B.P.	Specimen	J_{nrm}, A/m	χ $\times 10^{-5}$	Q_n	Q_l	J_{rs}, A m^2/kg	J_s, A m^2/kg	J_{rs}/J_s	T_c, °C
1a		8B794-2*	10.50	5.09	46	31	0.160	1.000	0.160	250,527
1b	260	9B131-5	5.94	2.06	64	44	0.136	0.683	0.199	95,525
1c		9B433-5	3.90	1.21	72	49	0.052	0.462	0.113	197,563
		9B437-5	3.34	1.17	64	43	0.042	0.398	0.106	207,574
		8B829-2*	17.80	3.54	122	84	0.293	0.974	0.301	131,487
2	440	8B833-4	22.90	3.78	135	101	0.301	0.875	0.344	143,526
		8B835-2*	26.00	5.44	107	79	0.426	1.470	0.290	468
		8B836-3*	20.60	3.05	151	112	0.293	0.904	0.324	107,476
3	700	1B475-2	5.39	1.97	61	49	0.117	0.561	0.209	72,536
		1B487-3	6.67	2.41	62	50	n.d.	n.d.	n.d.	110,550
4	760	1B755-1*	14.30	5.81	55	44	0.281	1.230	0.229	163,482
		1B757-3	21.10	7.00	67	54	n.d.	1.510	n.d.	260,497
		9B109-2*	12.60	2.48	113	98	0.337	1.060	0.318	124,472
5	1320	9B110-4	14.90	1.30	257	221	0.294	0.889	0.331	127
		9B113-4*	7.02	0.68	228	196	0.160	0.612	0.261	167
		9B117-2*	23.50	6.36	82	71	0.248	1.060	0.234	117,543
		1B730-1*	16.20	4.79	76	71	0.349	1.280	0.273	128,488
6	1690	1B732-2	27.20	7.44	82	75	n.d.	1.290	n.d.	180,486
		1B733-1*	17.90	4.07	98	91	0.448	1.360	0.329	146,483
		1B734-2	18.10	5.45	74	69	n.d.	1.210	n.d.	204
		8B889-4	10.00	9.15	24	18	0.126	1.050	0.120	554
7	2180	8B891-4	12.70	14.70	19	14	0.161	1.460	0.110	511
		8B896-2*	8.95	10.10	20	15	0.096	0.996	0.097	494
		8B897-4	23.30	12.80	41	30	0.226	1.190	0.190	531
8	2190	8B625-1	7.41	17.50	9	7	0.173	2.130	0.081	536
		8B631-1	4.43	22.90	4	3	0.276	2.600	0.106	517
		1B704-1*	16.40	4.62	79	65	0.394	1.300	0.303	172
9	2550	1B708-3	6.04	1.07	125	103	0.176	0.709	0.248	102,560
		1B710-1*	11.30	3.57	71	58	0.110	1.120	0.098	140,480
		1B714-3	23.50	6.15	85	70	n.d.	1.340	n.d.	257
		9B039-2*	6.42	6.76	21	15	0.082	0.882	0.093	182
		9B040-1	6.92	14.90	10	7	0.118	1.270	0.093	290
10	2890	9B041-3	6.76	13.10	12	8	0.149	1.510	0.099	282
		9B042-3	5.96	8.36	16	11	0.102	1.040	0.098	244
		9B046-3	16.50	8.04	46	32	0.255	1.150	0.222	184,551
11	3480	8B416-4	6.01	2.14	63	58	0.104	0.579	0.180	98
		8B417-1	12.30	3.41	80	55	0.207	0.839	0.247	240,515
		8B437-5	11.40	4.74	54	32	0.241	1.250	0.193	113,556
12	4050	8B439-2*	11.70	4.23	62	43	0.233	1.080	0.216	103,535
		8B440-2*	20.00	3.78	118	83	0.305	1.130	0.270	126,478
		9B445-3	9.80	7.34	30	29	0.161	0.947	0.170	85,564
		9B448-4	18.10	8.59	47	46	0.475	1.710	0.278	228,517
13	5160	9B449-2*	12.70	3.29	86	85	0.346	1.230	0.281	123,494
		9B451-2*	14.70	4.27	77	76	0.250	1.040	0.240	105,533
		9B454-4	14.20	17.80	18	18	0.417	2.270	0.184	480
		8B906-1	4.25	3.29	29	30	0.103	0.564	0.183	184,565
14	5650	8B907-2*	5.89	5.74	23	24	0.103	0.915	0.113	240,459
		8B910-3	5.57	6.63	19	20	0.200	1.370	0.146	197,467
15	6160	9B659-2*	6.39	3.46	41	42	0.168	0.854	0.197	78,540
		9B660-4	11.50	5.98	43	43	0.433	1.360	0.318	169,562
		9B961-2*	9.58	3.34	64	85	0.484	1.470	0.329	209
16	7300	9B962-3	7.23	3.40	48	63	0.356	1.140	0.312	186,505
		9B964-1	9.84	4.71	47	62	0.408	1.310	0.312	239
		9B965-2*	3.39	2.68	28	37	0.128	0.652	0.196	53,524
		9B666-3	5.14	2.20	52	60	0.215	0.835	0.258	571
17	7950	9B669-2*	10.10	5.99	38	44	0.182	1.020	0.178	93,539
		9B670-2*	5.80	3.94	33	38	0.065	0.594	0.110	80,548
		9B937-2*	1.50	0.66	51	40	0.030	0.368	0.081	79,504
		9B944-1	16.20	4.12	88	70	0.214	0.859	0.249	108,555
18	8740	9B945-3	9.32	4.22	49	39	0.301	1.240	0.242	266,513
		9B947-2*	6.12	3.10	44	35	0.154	0.832	0.182	87,536
		9B948-1	14.60	3.75	87	70	0.246	0.943	0.261	166,519

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TABLE 2. (continued)

Map (Figure 1)	Age, years B.P.	Specimen	J_{nrm}, A/m	χ $\times 10^{-5}$	Q_n	Q_l	J_{rs}, A m^2/kg	J_s, A m^2/kg	J_{rs}/J_s	T_c, °C
		9B949-2*	9.27	3.49	59	51	0.267	1.060	0.252	105,500
		9B950-3	16.30	6.09	60	52	0.450	1.430	0.315	201,518
19	9500	9B953-3	9.29	5.16	40	35	0.305	1.110	0.275	117,567
		9B955-2*	10.10	1.93	117	102	0.162	0.661	0.245	74,527
		9B959-3	18.50	4.75	87	75	0.438	1.240	0.353	177
		9B483-2*	9.39	5.34	39	39	0.082	0.761	0.108	83
		9B486-4	7.75	7.15	24	24	0.114	0.827	0.138	90,412
20	10,290	9B489-2*	9.26	3.24	64	63	0.316	1.240	0.255	193,510
		9B490-3	8.58	3.18	60	60	0.339	1.030	0.329	228,534
		9B492-4	6.22	3.45	40	40	0.189	0.842	0.225	91,553
		9B902-3	7.68	5.27	33	35	0.283	1.190	0.238	295,505
21	11,780	9B904-2*	4.12	1.10	84	89	0.199	0.807	0.247	237,493
		9B905-2*	8.64	3.95	49	52	0.186	0.879	0.212	150,548
		9B975-2*	10.80	3.84	63	59	0.307	1.060	0.290	120,545
22	13,210	9B976-1	6.83	4.50	34	33	0.213	0.925	0.230	140,572
		9B977-1	4.04	2.47	37	34	0.155	0.787	0.197	129,567
		9B984-2*	13.60	4.79	63	60	0.347	1.440	0.214	501

J_{nrm} is intensity of NRM; χ is volume susceptibility; Q_n is in situ Koenigsberger ratio ($J_{nrm}/\chi F$, where F = 35.7 μT); Q_l is laboratory Koenigsberger ratio [*Prévot et al.*, 1985]; J_{rs} is saturation remanence; J_s is saturation magnetization; T_c is Curie temperature; n.d. denotes quantity not determined.

*Magnetic properties and Thellier determinations conducted on adjacent specimens within the same core.

several percent higher than that acquired when the samples are cooled rapidly. One might expect therefore that slow cooling is preferable because it more closely approximates natural conditions, but these studies indicate that when a TRM is acquired by geologic or archeologic materials during slow cooling in nature, the Thellier method will yield estimates of the paleofield that are 10–20% too high. *Dodson and McClelland-Brown* [1980] caution, however, that the relationships they develop for magnetite and hematite may not be applicable to titaniferous materials with low Curie temperatures, such as contained in the majority of samples analyzed in this study. Moreover, *Halgedahl et al.* [1980] estimate that paleointensity errors would be less than 10% in the case of single-domain TRM carried by rapidly quenched specimens such as those from thin lava flows. Indeed, the fact that *Abokhodair* [1977], whose experiments were performed at the U.S. Geological Survey, obtained results that were accurate to within 7% of the known field indicates that our procedures are appropriate for the Hawaiian basalts.

Magnetic stability during the heating experiments was tested by redetermining the partial thermoremanent magnetization (PTRM) at a given low-temperature interval after the samples had been heated to successively higher temperatures (using the sliding PTRM-check method described by

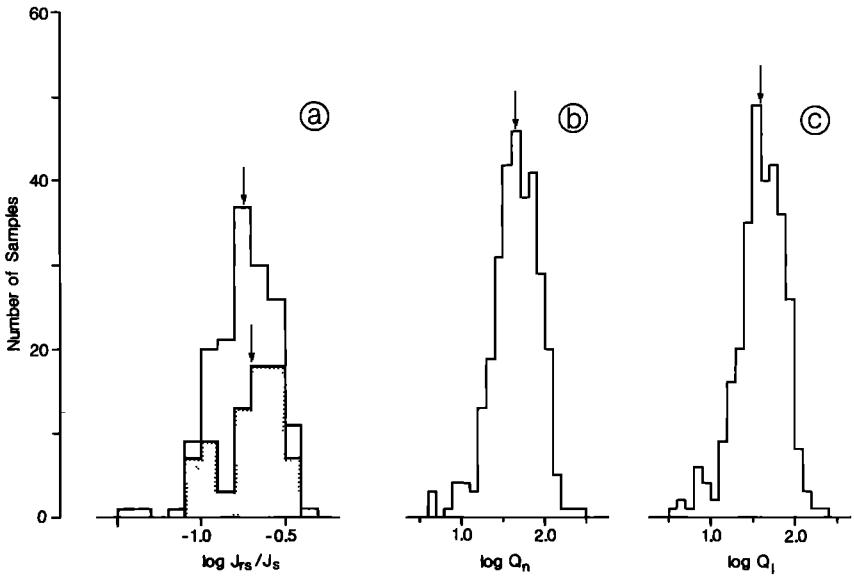


Fig. 4. Histograms showing the logarithmic distribution of (*a*) J_{rs}/J_s, (*b*) in situ Koenigsberger ratio Q_n, and (*c*) laboratory Koenigsberger ratio Q_l [*Prévot et al.*, 1985]. Arrows indicate mean values; the shaded area in Figure 4*a* corresponds to samples used in the paleointensity experiments.

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TABLE 3. Paleointensity Determinations for Hawaiian Holocene Lava Flows

Map (Figure 1)	Age, years B.P.	Sample	N	T_{mn}, °C	T_{mx}, °C	f	g	q	$F_e \pm \sigma(F_e)$, μT	$\overline{F}_e \pm$ s.d., μT	$\langle F_e \rangle$, μT	VDM, 10^{22} A m^2
1a		8B794-1	16	79	348	0.546	0.922	26.8	60.1 ± 1.1			
1b	260	9B131-5	7	38	169	0.333	0.790	5.9	55.9 ± 2.5	53.7 ± 5.3	52.6	10.92
1c		9B433-5	16	36	429	0.765	0.925	33.6	49.0 ± 1.0			
		9B437-5	16	59	465	0.824	0.926	36.4	49.6 ± 1.0			
		8B829-1	16	34	255	0.414	0.914	12.2	47.7 ± 1.5			
2	440	8B833-4	13	36	228	0.749	0.906	54.6	48.1 ± 0.6	47.6 ± 0.6	47.9	10.63
		8B835-1*										
		8B836-1	17	40	259	0.390	0.897	7.1	46.9 ± 2.3			
3	700	1B475-2	11	37	280	0.462	0.884	10.0	45.3 ± 1.9	44.5 ± 1.2	44.5	9.46
		1B487-3	8	38	131	0.420	0.818	8.0	43.6 ± 1.9			
4	760	1B755-2	19	52	370	0.699	0.935	47.6	44.8 ± 0.6	44.3 ± 0.8	44.1	10.59
		1B757-3	13	91	354	0.794	0.905	63.1	43.7 ± 0.5			
		9B109-1	17	40	255	0.734	0.926	46.3	38.4 ± 0.6			
5	1320	9B110-4	12	56	227	0.730	0.887	48.4	41.0 ± 0.6	42.4 ± 3.6	41.5	10.36
		9B113-3	13	55	259	0.652	0.884	44.2	43.1 ± 0.6			
		9B117-1	7	38	97	0.480	0.734	9.5	46.9 ± 1.7			
		1B730-2	12	67	247	0.574	0.903	34.0	42.4 ± 0.7			
6	1690	1B732-2	9	90	246	0.707	0.855	29.6	40.4 ± 0.8	39.3 ± 3.1	39.0	9.75
		1B733-2	16	57	248	0.816	0.924	61.9	39.1 ± 0.5			
		1B734-2	12	40	275	0.817	0.887	44.1	35.1 ± 0.6			
		8B889-4	13	49	482	0.472	0.876	17.5	72.1 ± 1.7			
7	2180	8B891-4*										
		8B896-3	19	45	560	0.809	0.906	47.4	46.3 ± 0.7			
		8B897-4	12	72	495	0.435	0.813	6.5	43.5 ± 2.4	51.1 ± 12.3	48.1	12.15
8	2190	8B625-1	17	49	540	0.903	0.874	62.9	41.8 ± 0.5			
		8B631-1	10	49	410	0.368	0.840	3.9	51.7 ± 4.1			
		1B704-2	15	77	312	0.746	0.921	71.0	45.1 ± 0.4			
9	2550	1B708-3	7	52	130	0.446	0.824	9.7	42.8 ± 1.6	42.8 ± 2.3	43.6	10.74
		1B710-2	14	61	292	0.600	0.908	23.5	39.7 ± 0.9			
		1B714-3	13	91	353	0.823	0.908	63.0	43.6 ± 0.5			
		9B039-1	13	51	192	0.564	0.902	15.3	62.4 ± 2.1			
		9B040-1	10	42	231	0.532	0.869	8.9	43.3 ± 2.3			
10	2890	9B041-3	10	42	231	0.547	0.864	11.3	46.8 ± 2.0	53.8 ± 11.6	51.7	12.42
		9B042-3	15	48	325	0.800	0.905	48.6	46.7 ± 0.7			
		9B046-3	12	132	399	0.486	0.866	11.4	69.7 ± 2.6			
11	3480	8B416-4	10	41	200	0.519	0.855	13.6	38.1 ± 1.3	45.4 ± 10.3	47.9	10.65
		8B417-1	15	30	372	0.668	0.903	34.9	52.7 ± 0.9			
		8B437-5	10	41	200	0.578	0.848	22.8	48.8 ± 1.1			
12	4050	8B439-1	7	57	114	0.442	0.786	16.7	52.2 ± 1.1	50.6 ± 1.7	50.6	11.53
		8B440-1	11	60	178	0.545	0.875	17.9	50.9 ± 1.3			
		9B445-3	10	40	200	0.477	0.853	17.4	35.2 ± 0.8			
		9B448-4	13	58	372	0.818	0.907	42.9	38.7 ± 0.7			
13	5160	9B449-1	13	60	238	0.529	0.897	19.2	31.8 ± 0.8	35.3 ± 2.6	36.1	8.57
		9B451-1	12	39	152	0.673	0.864	13.3	36.6 ± 1.6			
		9B454-4	6	198	401	0.514	0.794	6.6	34.1 ± 2.1			
		8B906-1	12	29	247	0.325	0.878	10.5	36.0 ± 1.0			
14	5650	8B907-1	16	51	331	0.617	0.909	32.4	36.5 ± 0.6	34.8 ± 2.5	33.9	7.71
		8B910-3	12	42	271	0.616	0.877	45.8	31.9 ± 0.4			
15	6160	9B659-1	5	60	100	0.161	0.726	2.5	44.2 ± 2.0	39.1 ± 7.3	35.2	6.44
		9B660-4	11	49	222	0.563	0.890	30.6	33.9 ± 0.6			
		9B961-1	12	100	288	0.621	0.890	29.9	24.4 ± 0.5			
16	7300	9B962-3	16	39	359	0.771	0.929	59.0	27.6 ± 0.3	26.5 ± 2.1	27.0	6.50
		9B964-1	15	73	345	0.763	0.920	50.3	28.8 ± 0.4			
		9B965-1	4	32	71	0.351	0.450	6.5	25.2 ± 0.6			
		9B666-3	6	97	376	0.293	0.794	4.6	30.8 ± 1.6			
17	7950	9B669-1	8	92	207	0.254	0.837	3.4	32.2 ± 2.0	31.0 ± 1.1	30.7	7.00
		9B670-1	7	91	188	0.297	0.827	7.8	30.0 ± 1.0			
		9B937-1	11	89	214	0.478	0.888	10.6	31.8 ± 1.3			
		9B944-1	10	43	198	0.750	0.854	49.6	47.3 ± 0.6			
18	8740	9B945-3	13	34	338	0.562	0.901	20.4	40.8 ± 1.0	41.1 ± 6.9	44.6	9.95
		9B947-1	9	72	192	0.277	0.857	4.0	37.5 ± 2.2			
		9B948-1	12	38	361	0.597	0.851	29.4	48.2 ± 0.8			

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TABLE 3. (continued)

Map (Figure 1)	Age, years B.P.	Sample	N	T_{mn}, °C	T_{mx}, °C	f	g	q	$F_e \pm \sigma(F_e)$, μT	$\overline{F}_e \pm$ s.d., μT	$\langle F_e \rangle$, μT	VDM, 10^{22} A m^2
		9B949-1	4	51	80	0.313	0.642	2.9	39.9 ± 2.7			
		9B950-3	8	48	161	0.611	0.853	14.8	34.8 ± 1.2			
19	9500	9B953-3	7	29	98	0.629	0.790	13.1	35.4 ± 1.3	39.9 ± 4.9	41.1	9.34
		9B955-1	11	62	211	0.480	0.863	11.6	43.7 ± 1.6			
		9B959-3	12	56	228	0.658	0.902	45.0	45.7 ± 0.6			
		9B483-1	6	61	116	0.240	0.782	2.2	51.8 ± 4.3			
		9B486-4	10	52	191	0.418	0.870	8.6	43.3 ± 1.8			
20	10,290	9B489-1	9	50	199	0.230	0.829	6.0	37.8 ± 1.2	41.4 ± 8.4	36.0	8.30
		9B490-3	10	29	198	0.175	0.851	2.1	44.7 ± 3.2			
		9B492-4	7	42	133	0.628	0.758	16.9	29.3 ± 0.8			
		9B902-3	12	95	355	0.313	0.870	10.6	30.0 ± 0.8			
21	11,780	9B904-1	15	78	330	0.396	0.909	22.7	37.2 ± 0.6	31.6 ± 5.0	33.4	8.58
		9B905-1	8	82	183	0.340	0.843	5.5	27.6 ± 1.5			
		9B975-1	12	34	184	0.569	0.901	13.9	31.8 ± 1.2			
22	13,210	9B976-1	12	38	361	0.589	0.892	25.3	36.5 ± 0.8	36.6 ± 3.4	37.9	6.96
		9B977-1	13	39	363	0.558	0.907	17.7	39.2 ± 1.1			
		9B984-1	15	199	553	0.951	0.883	131.7	38.8 ± 0.3			

N is the number of points in the $T_{mn} - T_{mx}$ interval; T_{mn} and T_{mx} are the minimum and maximum of the temperature range used to determine paleointensity; f, g, and q are the NRM fraction, gap factor, and quality factor, respectively [*Coe et al.*, 1978]; F_e is the paleointensity estimate for an individual specimen, and $\sigma(F_e)$ is its standard deviation [*Prévot et al.*, 1985]; $\overline{F}_e$ is the unweighted average paleointensity, with the plus and minus corresponding to the standard deviation; $\langle F_e \rangle$ is the weighted average paleointensity using the weighting factor fg/s of *Prévot et al.* [1985]; VDM is the virtual dipole moment [*Smith*, 1967].

*No good.

Prévot et al. [1985]). The method of *Coe et al.* [1984] was used to test whether appreciable amounts of chemical remanent magnetization (CRM) were acquired during the laboratory heatings. Weak field susceptibility was measured after each heating step for the last set of samples used in the paleointensity experiments to determine if changes in this property could be useful in detecting mineralogical changes that could affect the reliability of the results. Some degree of success using this technique was reported by *Tanaka* [1990], *Chauvin et al.* [1991], and *Tanaka and Kono* [1991].

The thermal decay of the thermoremanent magnetization (TRM) during the paleointensity experiments was found to mimic closely the thermal decay of saturation magnetization (J_s) during the strong field experiments (Figure 5). This result shows that the titanomagnetite populations within the samples are, indeed, discrete and have little overlap in their thermal characteristics. Note, however, the steepness of the low-temperature portion of the curve indicating a very narrow range of unblocking temperatures for the remanence carried by the high-Ti component. Such a precipitous loss of TRM with temperature made duplicating a given temperature for the second of a pair of heatings extremely critical because errors of only 2° or 3° would cause significant differences in the ratio of NRM loss to TRM gain. By not resetting either the cutoff temperatures or the oven voltages between each pair of field-off and field-on heatings, we were able to minimize but not totally eliminate this source of error. The large number of heating-cooling pairs (14 to 24), however, did enable us to distinguish between experimental noise and actual changes in NRM loss/TRM gain that could indicate magnetochemical changes due to heating. PTRM checks were much more difficult to control because both the cutoff temperatures and voltages had to be reset after heating to some higher temperature. Despite these difficulties, we show below that highly reliable paleointensity estimates apparently have been obtained from this suite of samples.

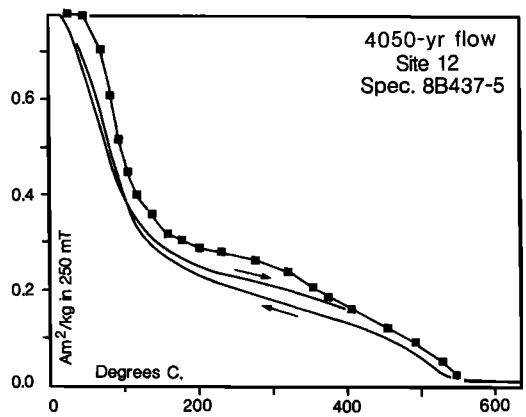


Fig. 5. Diagram showing the correspondence between thermal decay of both saturation magnetization (smooth curves) during the strong field, $J_s - T$, experiments and natural remanent magnetization (squares) during the paleointensity experiments. The NRM curve was normalized to the room temperature magnetization of the sample used in the $J_s - T$ experiment.

Data Analysis

Paleointensity data are best displayed on a NRM-TRM diagram [*Nagata et al.*, 1963], typical examples of which are shown in Figures 6–9. The NRM in these diagrams is the component directed along the stable NRM direction remaining after heating to a temperature T_i. The TRM plotted is the component projected onto the applied field direction that was acquired in the interval between T_i and room temperature. For an ideal sample, all such NRM/TRM points will fall on a straight line with the negative slope equal to the ratio of the ancient Earth's field to the applied laboratory field. In practice, however, only a fraction of the total NRM will

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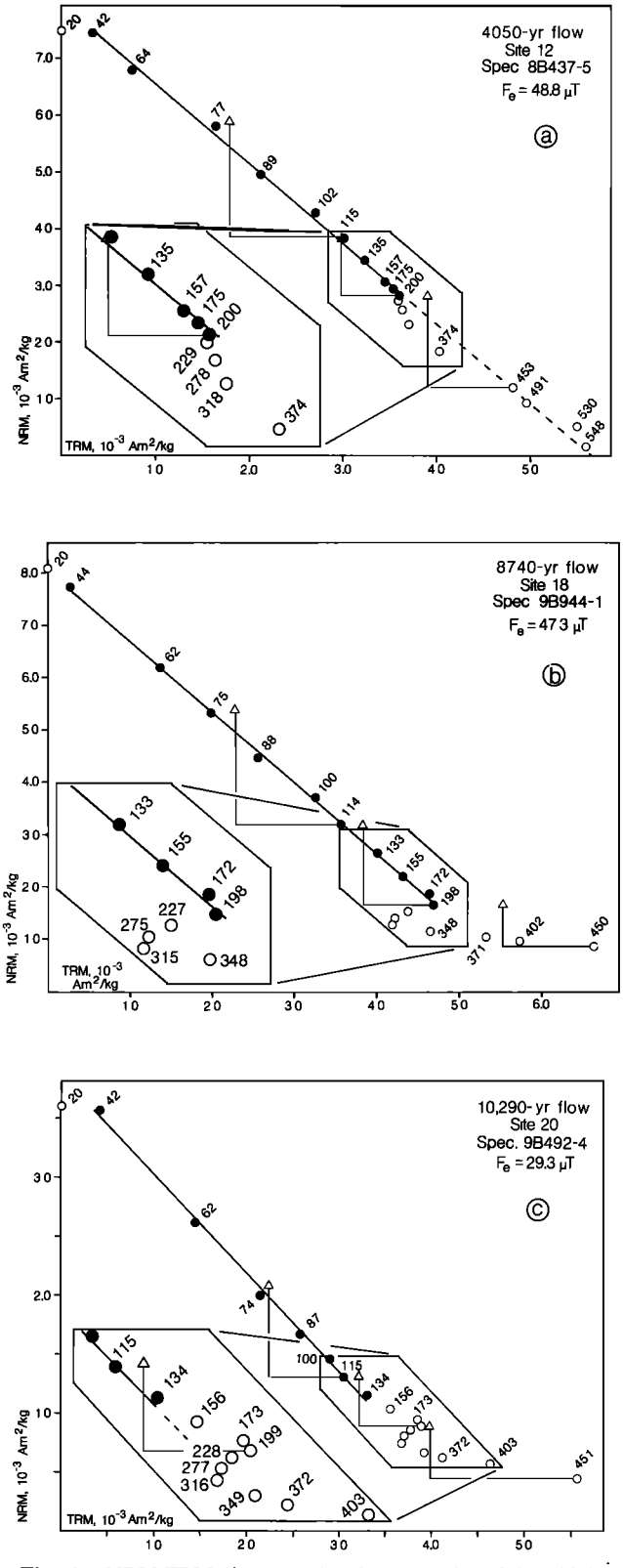


Fig. 6. NRM-TRM diagrams showing examples of the "kinks" that often signal the beginning of magnetochemical changes during the Thelliers' experiments. Circles are NRM-TRM points calculated for each double heating at the temperature indicated near each point. Lines connect each PTRM check (triangles) to the NRM-TRM point corresponding to the maximum temperature to which the specimen was heated before performing the check. Solid circles are NRM-TRM points used to determine the least squares line, shown as the heavy, solid line in the figures; open circles are points not used in the calculations.

yield reliable paleointensity values. Even rocks as young as these Hawaiian basalts can contain small amounts of viscous remanent magnetization (VRM) acquired in the present-day field, and, consequently, the room temperature measurement often will not fall on the linear part of the NRM-TRM diagram (Figures 6–9).

Magnetochemical changes occurring during the laboratory heatings often are characterized by loss of NRM accompanied by a pronounced change in the ability of the sample to gain TRM as shown by the "kinks" in Figure 6. The sample in Figure 6*a* is unusual in that three of the four points after the kink fall on a continuation of the least squares line through the lower-temperature NRM-TRM pairs. It is much more common for NRM-TRM points to display a progressive flattening of slope following the kink as shown by the samples in Figures 6*b* and 6*c*. Occasionally, the NRM-TRM points began to deviate from linearity before the kink occurred (Figure 6*c*). In this last example, note that reasonable PTRM checks were obtained throughout the experiment. Magnetochemical changes were more difficult to detect in other cases, but alteration eventually did occur at some point in the heating cycle for all samples in our collection. Consequently, the NRM-TRM points defining a straight line were obtained from 16% to 93% of the initial NRM intensity. Because of this alteration upon heating, none of our samples would be suitable for any paleointensity method that required an initial heating to above their Curie temperatures.

Paleointensity experiments were performed on 79 samples and only 2 of those analyzed failed to yield an estimate of the paleofield (Table 3). These rocks therefore seem to be ideally suited for paleointensity analysis by the Thelliers' method. Unfortunately, there is currently no consistency test that can ensure the accuracy of a given paleointensity and the factors listed below, considered as a group, were used to assess the reliability of our estimates.

Linearity of the NRM-TRM points. The first requirement for obtaining a paleointensity estimate is, of course, that the NRM-TRM points within a given range of unblocking temperatures fall on a straight line. We follow the suggestion of *Coe et al.* [1978] and consider only those line segments that are defined by at least four points, and those points must span a range of at least 15% of the total NRM intensity. This last condition is used to avoid interpreting those data that show excessive systematic curvature such as those obtained from samples containing MD magnetites [*Levi*, 1977]. Linearity, however, cannot establish the reliability of the paleointensity estimate in the absence of additional supporting data. Numerous examples have been published of NRM-TRM points that define a straight line even when chemical alteration can be demonstrated, and more than one straight-line segment is often present in the data obtained from a single specimen. Linearity of NRM-TRM points commonly is observed at the lowest temperature steps during the removal of secondary components of magnetization such as VRM and isothermal remanent magnetization (IRM) and, again, at the highest-temperature steps when chemical alteration is occurring. Because our samples are so young, secondary components are minimal and rarely persist past the first or second heating pairs. Similarly, fortuitous straight-line segments resulting from chemical alteration often followed an interval where the capacity to acquire a TRM was greatly diminished. These factors made it fairly unambiguous as to which segment of the NRM-TRM dia-

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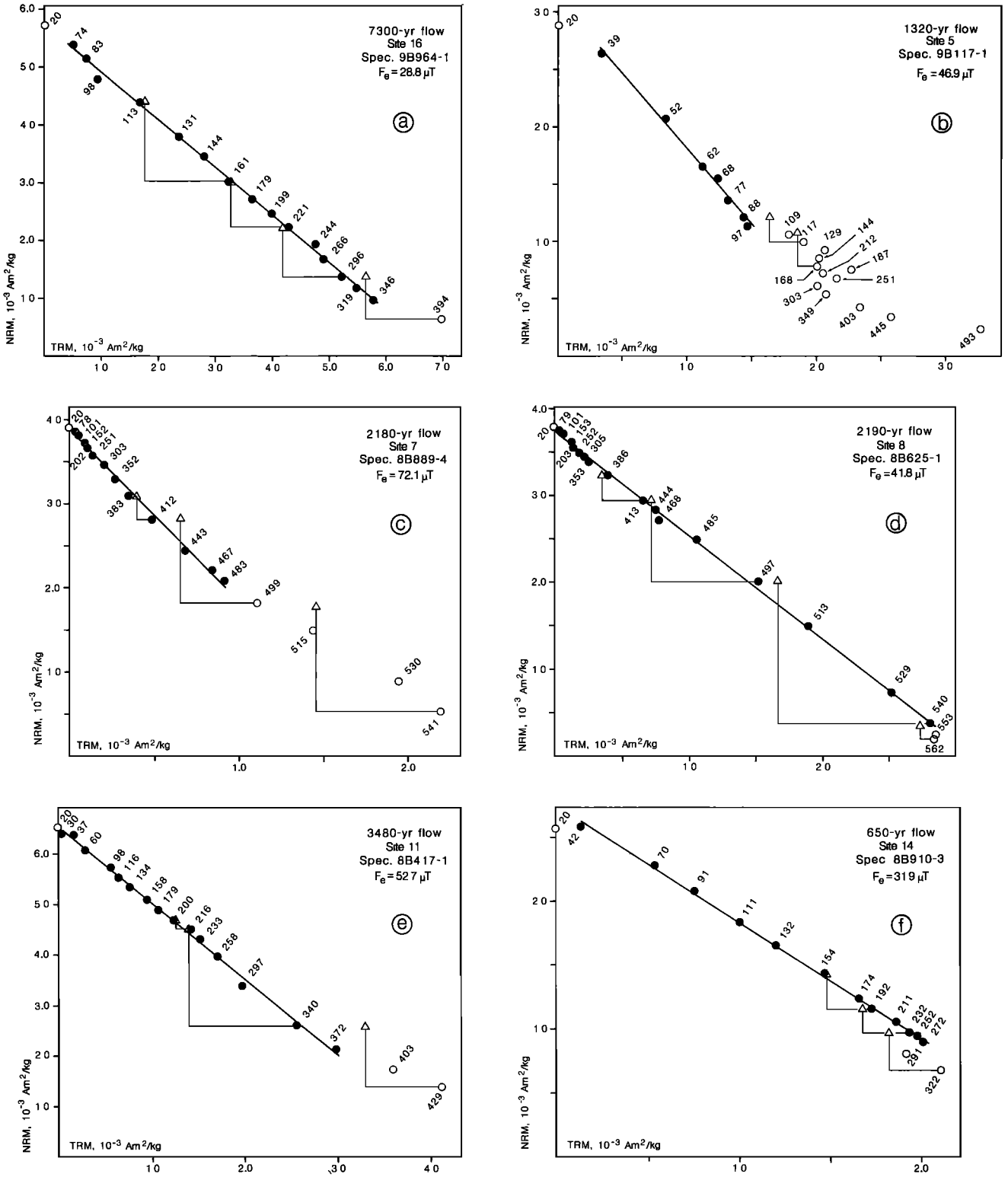


Fig. 7. Typical examples of paleointensity data (NRM-TRM diagrams) from the Hawaian basalt flows. See Figure 6 for additional explanation.

gram should be interpreted for many of our samples. Because a recently acquired VRM can be quite close to the TRM direction and therefore difficult to detect, the initial room temperature measurement was not included in any of the paleointensity estimates even if it fell on the same line as subsequent NRM-TRM points. The actual paleointensity value calculated will be unaffected by this decision, but the statistical parameters given in Table 3 will be more conservative.

PTRM checks. The PTRM check is the most commonly used method to test the reliability of paleointensity determinations. PTRM checks for a number of our specimens

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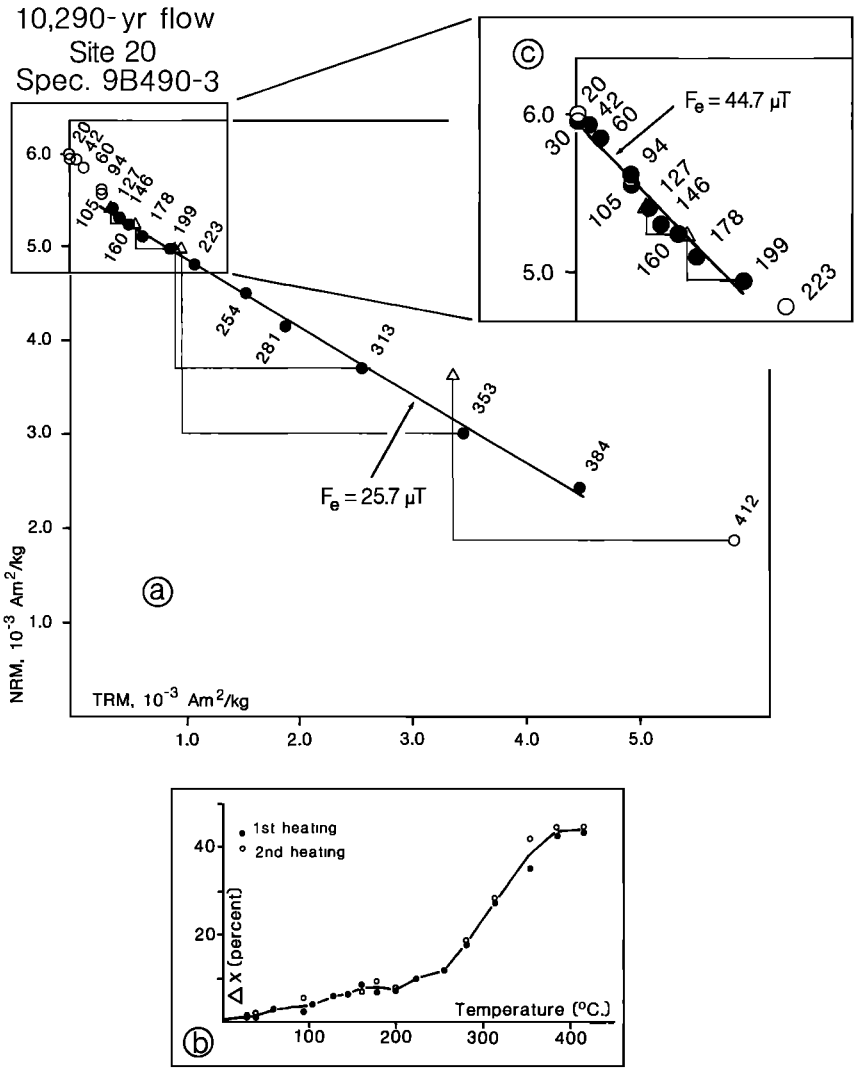


Fig. 8. (*a*) NRM-TRM diagram apparently yielding a high-quality paleointensity estimate (f = 0.534, g = 0.846, q = 25.6, and four positive PTRM checks). A marked change in low-field susceptibility, however, indicates (*b*) that chemical alteration is occurring during this interval and (*c*) that the lower-temperature linear segment is the correct portion of the NRM-TRM curve to interpret. Symbols as in Figure 6.

differed slightly, but unsystematically, from the least squares line, and paleointensity estimates in those cases seemed to be unaffected when compared to other specimens from the same flow. We believe these deviations represent experimental noise because of the difficulty in controlling PTRM checks and consider this test to be positive (i.e., no chemical alteration indicated) if it yielded a value that was no more than about 5% of the value previously determined for the same temperature, particularly at higher temperatures where the test included a substantial portion of the TRM [e.g., *Coe*, 1967]. Progressive or systematic changes in PTRM, regardless of amount, were considered to be a more significant indication that the TRM spectrum had been altered. The PTRM checks shown in Figures 6–9 are typical of behavior for all specimens. When a temperature of a previous heating step was not duplicated exactly, the NRM value for the PTRM check was obtained by linear interpolation between the bracketing heating steps. Systematic deviations from the straight line, when present, usually began after a significant break in slope of the NRM-TRM curve was noted. As is the case with other consistency tests, however, PTRM checks cannot provide indisputable evidence regarding the presence or absence of alteration [*Coe et al.*, 1978; *Grommé et al.*, 1979; *Prévot et al.*, 1983; *Perrin et al.*, 1991].

CRM acquisition. Because of the small magnitudes of secondary components of NRM present in our rocks, chemical changes during the laboratory heatings are probably the largest sources of error in the paleointensity determinations. A CRM acquired during heating can often be detected by a systematic displacement of the NRM direction toward the applied field direction if the NRM and applied field are not parallel [*Coe et al.*, 1978, 1984]. We used the method of *Coe et al.* [1984] to analyze the data obtained from each specimen and found that the potential for error R that the CRM acquired would cause in the paleointensity estimates ranged from less than 1% to a maximum of 11% in the linear segments analyzed. Angular deflections toward the applied field direction were less than 5°.

Low-field susceptibility. Low-field susceptibility was measured after each heating step only during our last set of

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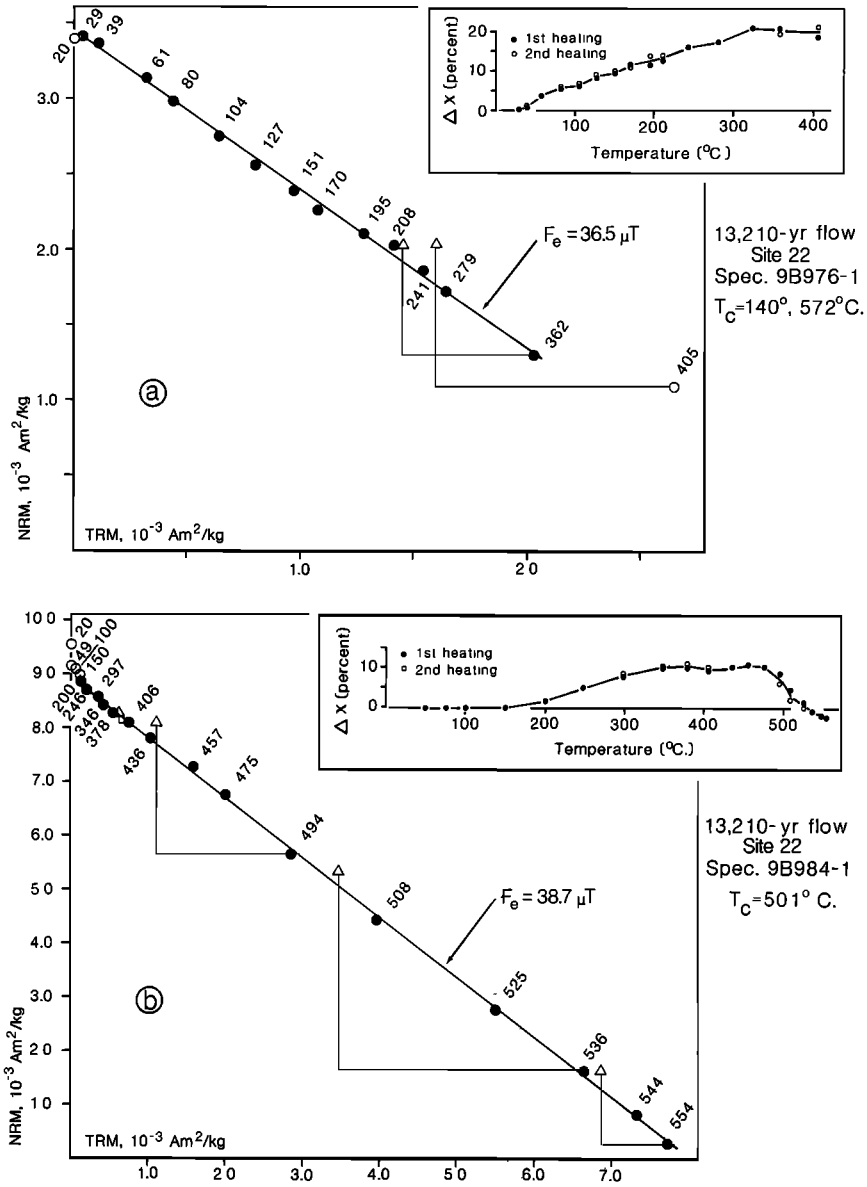


Fig. 9. NRM-TRM diagrams showing paleointensity estimates obtained from (*a*) low-temperature and (*b*) high-temperature titanomagnetite populations from the 13,210 year B.P. flow (site 22). Symbols as in Figure 6.

experiments (30 samples), and we are unable, therefore, to make many generalizations about the applicability of this property as a consistency check for the rocks in our collection. It was obvious, however, that large changes in susceptibility can occur without any apparent effect on the linearity of the NRM-TRM diagram as first noted by *Coe* [1967] and more recently by *Tanaka* [1990], *Chauvin et al.* [1991], and *Tanaka and Kono* [1991]. Susceptibility changes often occurred at temperatures as low as 30°C, and increases up to as much as 20% seemed to have no effect on the remanence. Because these initial changes in susceptibility do not seem to affect the paleointensity, they must represent changes in phases other than those carrying the remanence. *Chauvin et al.* [1991] also speculate that some of these increases in weak field susceptibility may be due to a stress effect rather than indicating mineralogical changes. In any case, sudden deviations in the rate of change seemed to be more important than total amount as noted by *Tanaka and Kono* [1991] and shown in Figure 8. Data for this sample from the 10,290 year B.P. flow (site 21) display a linear segment spanning 53% of the NRM between 127° and 384°C, and four positive PTRM checks are contained within this interval (Figure 8*a*). Weak field susceptibility measurements, however, indicated that major changes occurred during the same temperature interval (Figure 8*b*) and that the linear segment between 30° and 199°C (Figure 8*c*) was the correct portion of the NRM-TRM curve to interpret. This low-temperature segment provides a paleointensity estimate that seems most reasonable when compared with other estimates from the same flow and others in the same age range (Table 3). Abrupt changes in susceptibility, if any, for the rest of the samples occurred at about the same temperature where there were obvious breaks in the slope of the NRM-TRM curve and a systematic deviation of the PTRM checks from the straight line.

Statistical parameters. The overall quality of our paleointensity estimates can be evaluated using the statistical

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parameters developed by *Coe et al.* [1978]. In Table 3, T_{mn} is the temperature above which secondary magnetizations are negligible and T_{mx} is the temperature above which magnetochemical changes occur. The NRM fraction f is the percentage of the total NRM represented by the linear portion of the NRM-TRM curve between T_{mn} and T_{mx}. The NRM fraction in these experiments range from 16 to 93%, with values typically about 50–60%. The gap factor g is a measure of how evenly the NRM-TRM points are distributed along the straight-line segment. The more irregular are the widths of the gaps between points on the segment being analyzed, the more g will decrease from unity. It is obvious from g values given in Table 3 and a visual inspection of Figures 6–9 that significant errors in slope due to inadequate sampling of the NRM spectrum are unlikely. Factors f and g are combined with a measure of the scatter of NRM-TRM about the straight-line segment to produce an overall index of quality q. Quality factors (Table 3) are consistently above 10 and are indicative of analytically precise data.

Within-flow consistency. Most of the paleointensity determinations for a given flow are remarkably consistent (Table 3) and attest to the reliability of the data unless some systematic and subtle magnetochemical changes affect our experiments. Four of the flows studied had one sample with a single high Curie temperature component in addition to samples with low to intermediate Curie temperature components. For the flows at sites 13, 17, and 22 (5160, 7950, and 13,210 years B.P., respectively), both low- and high-Curie-temperature populations yielded similar paleointensity values (Figure 9). Note, in Figure 9*b*, that the indicated Curie temperature is 501°C, whereas the NRM-TRM diagram is linear to 554°C. This discrepancy arises because the magnetic properties and Thellier experiments were performed on adjacent specimens within the same core before the desirability of having both experiments on the same specimen was realized. The 501°C Curie temperature is probably not typical of the core as a whole. The high-Curie-temperature sample from the fourth flow (site 2, 440 years B.P.) failed to yield an interpretable NRM-TRM diagram. Concordant results from two titanomagnetite populations within a single flow are a strong indication that no insidious alteration is occurring because both phases would not be expected to react in the same manner.

Results

Paleointensity estimates and associated statistical parameters are given in Table 3. Because significant differences in the measures of quality are commonly observed for samples within a single unit, both unweighted and weighted mean paleointensities were determined and the virtual dipole moment (VDM) was calculated from the weighted mean. Weighted means were calculated using the weighting factor defined by *Prévot et al.* [1985], and these are shown in Figure 10. Because the weighting factor used differs from the weight derived from the theory of errors and cannot therefore be used to calculate a weighted standard error, we follow *Prévot et al.* [1985] and express uncertainties in the paleointensity estimate by using the standard deviation about the unweighted mean. Included in Figure 10 are the data from *Tanaka and Kono* [1991] and those data from *Coe et al.* [1978] that are well constrained in age. Of the flows studied by *Coe et al.* [1978], only minimum ages are available for what they called the Maniania Pali II and Hilina Pali II flows and bounding ages for the Puu Loa Loa flow; these are not shown in Figure 10. Also not shown is the result for the Visitor Center flow whose age is questionable (see above). Age uncertainties shown about each datum are comparable to those given in Table 1.

Experience has shown that substantial differences in apparent paleointensity can occur within a single lava flow, and we attempted therefore to obtain a minimum of three determinations for each unit. When this was not possible because of the lack of suitable samples, another flow with a statistically indistinguishable age and remanence direction was selected under the assumption that both recorded the same instantaneous (geologically) geomagnetic field. The only exception made was for the flows erupted at 760 and 700 years B.P. (sites 3 and 4), whose mean directions differed by 17.6°. These flows yield statistically indistinguishable ages and paleointensities (Tables 1 and 3), but the large difference in remanence directions indicates an age difference of the order of centuries. Consequently, both paleointensities are plotted at the extremes of their age uncertainties in Figure 10. We consider the two flows to be in their correct stratigraphic position (also see the secular variation curves of *Holcomb et al.* [1986]). Only two samples were available from the 6160 year B.P. flow at site 15, and no additional flows with a similar age have been sampled.

When all the factors used to evaluate the paleointensity determinations (listed above) are considered, our estimates of the paleofield, in general, seem to be quite reliable. It was not possible to calculate a paleointensity for only two of the 79 specimens analyzed and both of these (specimens 8B835-1 and 8B891-4 in Table 3) were high-T_c samples that yielded $J_s - T$ curves whose heating and cooling curves differed in shape (Figure 2*h*). Note also that many of the data seem to be serially correlated (Figure 10), which would not be expected if serious errors were being introduced in the experiments.

Some of the low-Curie-temperature samples proved to be challenging because the rapidity at which the NRM intensity was lost made it difficult to obtain many NRM-TRM pairs before alteration began. Specimen 8B437-5 from the 4050 year B.P. flow (site 12), which contains titanomagnetite with a Curie temperature of 113°C (Table 2 and Figure 5), was heated to 200°C before it began to alter, and 10 points could be used to estimate the paleofield (Figure 6*a*). Other specimens such as 9B117-1 from the 1320 year B.P. flow (site 22), with a nearly identical high-Ti phase (T_c = 117°C), clearly had altered after heating beyond 100°C as attested to by the deviation from a straight line and the poor PTRM checks (Figure 7*b*). Only seven NRM-TRM pairs could be used for this determination, and the rapid loss in NRM intensity between 40° and 60°C resulted in large gaps in the record and relatively low measures of quality (Table 3). Despite the poorer quality of the data from this sample compared with others from the same flow, the paleointensity estimates are very similar. Other such examples can be found in Table 3.

The flows at sites 7, 8, 10, and 20 (2180/2190, 2890, and 10,290 years B.P., respectively) have considerable variation in the paleointensities indicated by different samples within the same unit. It is not uncommon for such variation to occur, as noted above, but the cause of the discrepancies is not obvious. For the 10,290 year B.P. flow (site 20), we showed that an anomalously low paleointensity could be

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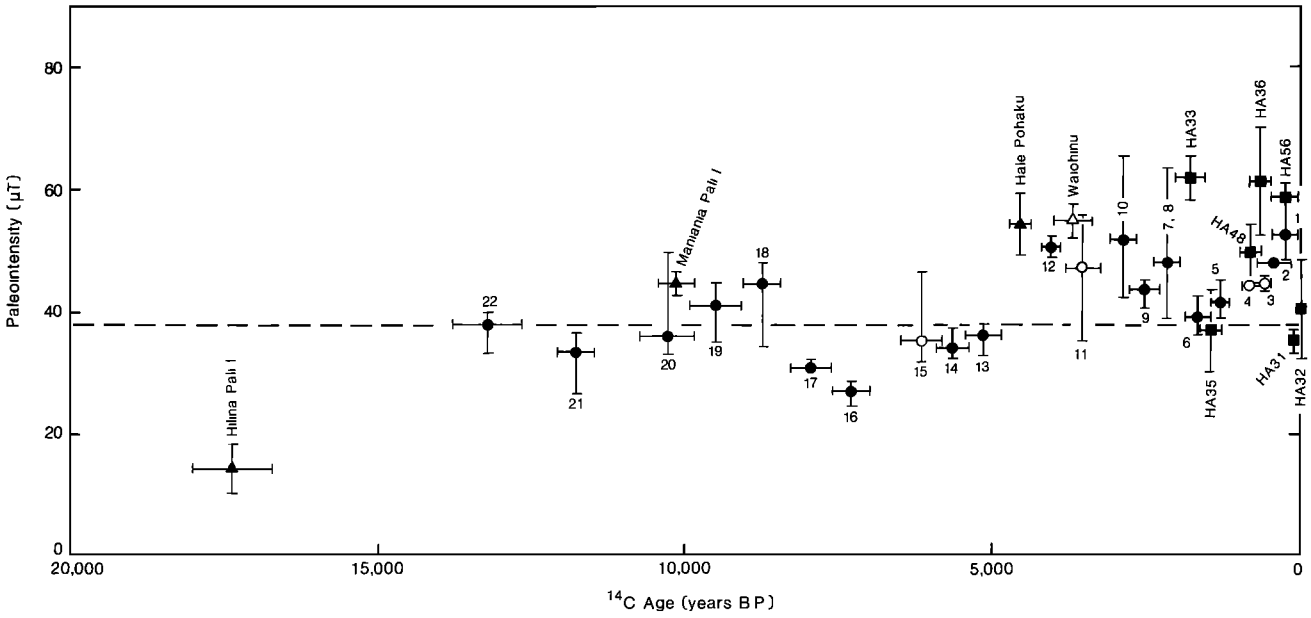


Fig. 10. Paleointensity as a function of age for the Hawaiian ^{14}C-dated lava flows. Circles are weighted means from Table 3; triangles are data from *Coe et al.* [1978]; squares are data from *Tanaka and Kono* [1991]. Labels are flow designations used by the authors. Open symbols are values based on fewer than three determinations. Uncertainties in age as in Table 1; uncertainties in paleointensity are standard deviations about the unweighted means (see text). Dashed horizontal line is the present field intensity in Hawaii.

obtained for specimen 9B490-3 if the subtle magnetochemical changes that were occurring were not recognized (Figure 8). However, the same problem clearly cannot account for the low paleointensity indicated by specimen 9B492-4 from the same flow (Figure 6*c*). All samples from this flow have a large low-Curie-temperature component, indicating that the variance is not mineralogically controlled. In the case of combined site 7/8 (2180/2190 year) which has the largest within-unit variation, all samples have been deuterically oxidized, and no low-Curie-temperature component is present (Table 2). There seems to be no reason to suspect the data from the sample providing either the highest (Figure 7*c*) or lowest (Figure 7*d*) paleointensity estimate. It is assumed that by studying several samples from each unit and calculating a weighted mean paleointensity, a reasonable estimate of the paleofield will be obtained. For example, four of the five reliable samples from these sites yield paleointensities ranging from 41.8 to 51.7 μT, the fifth indicates a 72.1-μT field, and the weighted mean paleointensity of 48.1 μT seems reasonable when compared with some of our other units from this time period (Figure 10). We note, however, that *Tanaka and Kono* [1991] obtained a substantially higher paleointensity of 61.8 μT for a flow (HA33) with a ^{14}C age of 1810 years B.P. There now seems to be a distinct possibility (J. P. Lockwood, oral communication, 1992) that the charcoal yielding the 1810-year age was obtained from beneath the flow overlying the one sampled for paleomagnetic study by *Tanaka and Kono* [1991]. If true, flow HA33 could be significantly older than thought, and its paleointensity could be more compatible with our results. Additional flows erupted at about 2000 years B.P. need to be analyzed to provide a better estimate of the paleofield at that time.

Samples from the 260-year flow (sites 1a, 1b, 1c) have undergone considerable high-temperature oxidation and it was difficult to find samples to satisfy our requirement that greater than 50% of the remanence should be carried by the low-Curie-temperature component. For this reason, samples from three separate outcrops were needed to determine the paleointensity for this unit. *Tanaka and Kono* [1991] sampled the same unit at a fourth locality (HA56) and obtained a value (58.8 μT) that is 9% higher than the unweighted mean that we have determined (53.7 μT) and consistent with the within-site dispersion of about 10% that they estimated. An overall within-site dispersion of about 10% is typical also of our data set (Table 3).

Paleointensities determined by *Tanaka and Kono* [1991] for flows erupted at 1470 and 830 years B.P. (HA35 and HA48, respectively) are quite similar to values we obtain for the same periods of time (Figure 10). The 61.3-μT value they obtain for the 670 year B.P. flow (HA36) seems anomalous, although this unit does have a large within-site dispersion. Considered as a whole, however, the paleofield in Hawaii during the interval between 1000 and 200 years B.P. seems to have been about 45% stronger than it is at the present time. Because of this persistence of strong paleofields, the strong paleointensity (58.0 μT) determined for the Visitor Center flow [*Coe et al.*, 1978] does not preclude the possibility that it erupted as recently as 500 years ago (see above).

SUMMARY AND DISCUSSION

All data shown in Figure 10 have been converted to their equivalent virtual dipole moments and plotted in Figure 11 to facilitate comparison with data from other regions. Uncertainties about each VDM are expressed as standard deviations about the unweighted mean, as was done for the field values in Figure 10. *McElhinny and Senanayake* [1982] analyzed 1167 worldwide archeomagnetic results and calculated 500-year average dipole moments back to 4000 years B.P. and 1000-year averages for earlier periods. These data,

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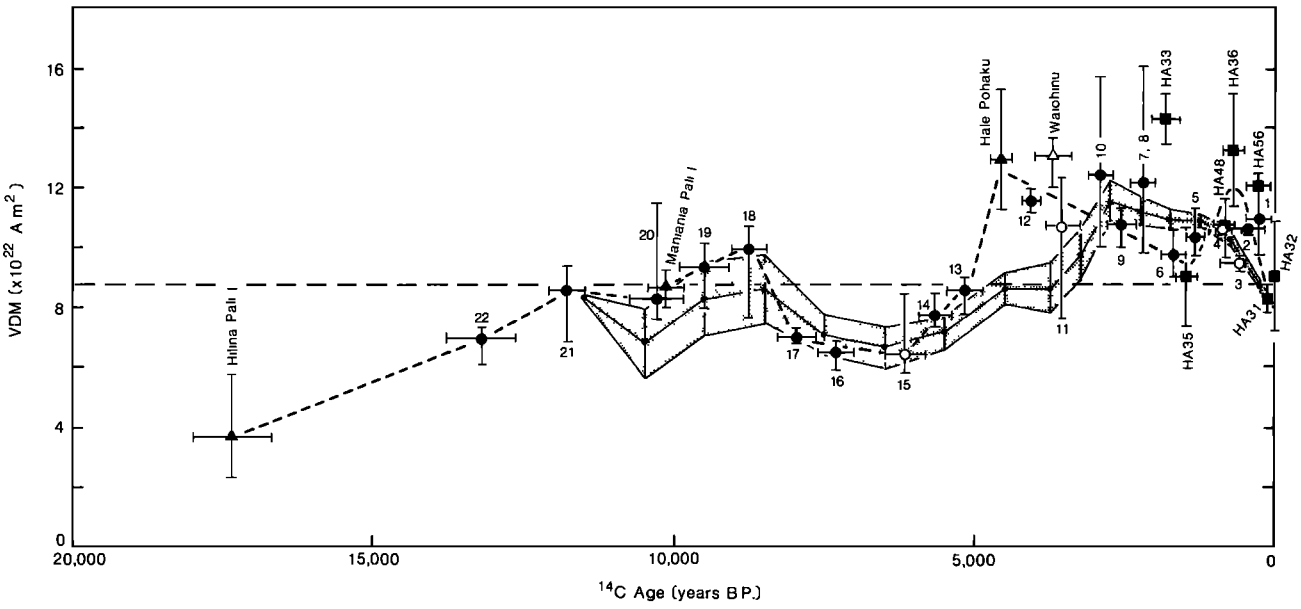


Fig. 11. Virtual dipole moments (VDMs) as a function of age for the Hawaiian ^{14}C-dated lava flows. Dashed curve is a rough fit (by eye) of all paleointensity data from Hawaii (see text). Global-mean dipole variation curve [*McElhinny and Senanayake*, 1982] for Holocene time is shown with its 95% confidence envelope (shaded area). Dashed horizontal line is the worldwide average VDM for the past 10,000 years [*McElhinny and Senanayake*, 1982]. Symbols as in Figure 10.

shown in Figure 11 with their 95% confidence envelopes (shaded area), provide an estimate of dipole intensity fluctuations back to about 12,000 years B.P. Readers should be cautioned, however, that despite data becoming available from different parts of the world, this mean curve remains heavily biased toward the European region where many of the earliest paleointensity determinations were made.

The heavy, dashed curve in Figure 11 is an attempt to roughly fit (by eye) our data and those of *Coe et al.* [1978] and *Tanaka and Kono* [1991]. Because we obtained only two determinations for the 700 year B.P. flow (site 3), we give more weight to *Tanaka and Kono*'s [1991] value (based on five determinations) for the 670 year B.P. flow (HA36) and show a broader peak in intensity during the 700- to 500-year interval than would be indicated solely by our data. Following the intensity high near 700 years B.P., we show the curve in Figure 11 as descending through our 440- and 260-year data (sites 2 and 1), the data for the A.D. 1840 historic flow (HA31 [*Tanaka and Kono*, 1991]), and ending at the present-day value. *Tanaka and Kono* [1991], however, obtained a result for the A.D. 1960 historic lava (HA32) that is 15% higher than the expected field, although the within-site dispersion is quite high (20%) when all the data from this flow are considered. *Abokhodair* [1977] also studied the A.D. 1960 flow and obtained results (47.0 μT in air; 42.3 μT in vacuum) that are in general agreement with the 40.3-μT value obtained by *Tanaka and Kono* [1991]. The A.D. 1960 flow was one of only two units out of the 16 historic flows studied by *Abokhodair* [1977] that yielded anomalous results when compared with reference values for Hawaii. Abokhodair considered the data from this flow to be unreliable because the estimates were obtained from temperature intervals containing less than 10% of the total NRM. He also noted that polished sections of samples from this flow revealed highly oxidized titanomagnetite grains with partially resorbed edges enclosed in fresh titanomagnetite shells. Perhaps the anomalous values could be a result of some type of interaction between the different titanomagnetite generations. For whatever reason, results from the A.D. 1960 flow seem atypical of the historic flows and should be viewed with caution.

Nondipole Field

The paleointensities determined by *Coe et al.* [1978] were consistently above the mean archeomagnetic curve that was known at that time, leading them to conclude that significant nondipole sources existed in the vicinity of Hawaii during the interval between 4600 and 2600 years B.P. Comparing the curve of intensity variations for the Hawaiian lava flows with the more recent worldwide dipole variation curve in Figure 11 indicates that nondipole sources were, indeed, present from about 5000 years B.P. to perhaps as recently as the past 200 years. The close correspondence between the two curves during earlier periods, however, indicates that the nondipole field must have been virtually absent during the interval between about 12,000 years to 5000 years B.P. just as it is at the present time.

Although the paleointensity variations are indicative of nondipole sources, the nondipole field does not appear to be reflected in the paleomagnetic directions. *Doell and Cox* [1972] calculated an angular dispersion of VGPs for Hawaii (10.8° with respect to the geographic axis) that was clearly anomalously low when compared with worldwide averages and led them to suggest that the nondipole field was greatly subdued in the central Pacific during the past 0.7 m.y. It was realized later that this dispersion could be biased toward a low value because two of the thick volcanic series sampled (Puna and Kau) probably represented short periods of time. *Coe et al.* [1978] combined data from their ^{14}C-dated lava

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flows with *Doell and Cox*'s [1972] data, exclusive of these two series and the historic lavas, and calculated 12.7° of angular dispersion with respect to the mean of all VGP's for the Brunhes Chron. This dispersion agreed well with predicted values and indicated normal secular variation during the Brunhes. *McWilliams et al.* [1982] also reanalyzed the *Doell and Cox* [1972] data set, combined these with results from their more extensive ^{14}C collection and calculated 11.2° of dispersion that, again, is anomalously low. An attempt was made to thin the ^{14}C data using different, nonoverlapping time windows, but the data set remained strongly biased toward ages younger than 6000 years [*McWilliams et al.*, 1982, Figure 1], and the calculated dispersion probably is more representative of Holocene time rather than the entire Brunhes. Recently, *Peng and King* [1992] used a continuous paleomagnetic record for the last 13,000 years, contained in a piston core form Lake Waiau on Mauna Kea Volcano, and calculated 5.9° of dispersion with respect to the mean VGP and 9.1° with respect to the geographic axis. Both values are much lower than predicted by global models and support the hypothesis of anomalously low secular variation. The values calculated by *McWilliams et al.* [1982] and *Peng and King* [1992] again raise the possibility that geomagnetic secular variation at Hawaii was manifested entirely by changes in field intensity (see discussion by *Coe et al.* [1978]). The fact that mean VGPs calculated by *McWilliams et al.* [1982] and *Peng and King* [1992] are very similar and differ significantly from the geographic pole, however, could be interpreted as resulting from incomplete averaging of geomagnetic secular variation, especially when considered along with the very low dispersion figures. On the other hand, when measured in other parts of the world, the time-averaged field over 10^4 years has been shown to be that of a geocentric axial dipole [e.g., *Lund and Banerjee*, 1985; *Verosub et al.*, 1986], and *Peng and King* [1992] consider that some persistent feature of the regional geomagnetic field is the most likely cause of the shallow average magnetic inclinations recorded in their sedimentary records and by the lava flow record of *McWilliams et al.* [1982].

Accuracy of the Radiocarbon Time Scale

A recent study of coral cores obtained near Barbados shows that ^{14}C and U-Th dating methods applied to this material yield concordant ages to about 9000 years B.P. but differ systematically prior to that time [*Bard et al.*, 1990]. The ^{14}C ages proved to be consistently younger than those obtained by U-Th, with the maximum difference being about 3500 years at 20,000 years B.P. *Bard et al.* [1990] argue that although neither method is well calibrated beyond 9000 or 10,000 years, the U-Th ages seem most reliable when compared with dendrochronology, varve dating, and paleomagnetism. If true, the difference between the two methods is equivalent to a 400% change in atmospheric $^{14}C/C$, a change so large that the authors consider that it only can be due to a significant decrease in geomagnetic intensity during the last glacial period.

Unfortunately, few paleointensity determinations are available for testing this hypothesis. Among paleointensity data from the Hawaiian lavas, a VDM calculated for the Hilina Pali I flow that erupted at 17,360 years B.P. [*Coe et al.*, 1978] is nearly 60% lower than the average VDM calculated by *McElhinny and Senanayake* [1982] for the past 10,000 years. *McElhinny and Senanayake* also calculated a mean VDM for 14 data (including those from the Hawaiian lavas) from the period between 50,000 and 15,000 years B.P. and found a field intensity about one-half the value for the past 10,000 years. This last analysis excluded the four strong paleointensities determined by *Barbetti and Flude* [1979] for samples from a 19,420 year B.P. fireplace in southeastern Australia because the values are associated with the Lake Mungo excursion and may not be typical of the average field at that time. On the other hand, 24 paleointensity determinations on lava flows erupted during the 46,600-year-old Laschamp excursion both in France [*Chauvin et al.*, 1989] and Iceland [*Marshall et al.*, 1988; *Levi et al.*, 1990] are consistently lower than the worldwide average by 75% or more. Determination of relative paleointensity from sedimentary records remains controversial; however, the records contained by marine cores from Ontong-Java plateau [*Constable and Tauxe*, 1987], the North Atlantic [*Tauxe and Valet*, 1989], and the Tyrrhenian and eastern Mediterranean Seas [*Mazaud et al.*, 1991] show broad similarities to the worldwide archeomagnetic record, including very low intensities between about 20,000 to 15,000 years B.P. Thus, although data are sparse, there are indications that the last glacial period was marked by some very low paleointensities. If these indications prove correct, the available data from Hawaii may place the younger limit to the period of low paleointensities in the interval between about 17,000 and 13,000 years B.P. These ages will have to be increased if a prolonged low intensity interval did occur, but their relative ages should remain approximately correct. Lava flows erupted during the interval between about 30,000 and 10,000 years B.P. are currently being investigated to provide additional data from this critical period.

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